

LEARNING RESOURCE GUIDE

ISC2 CCSP: Certified Cloud Security Professional (Aug 2026) | Professional Participant Workbook

Learning Objectives

By the end of this lecture and its accompanying guide, you will be able to:

- By the end of this lecture, you will be able to distinguish between Type 1 and Type 2 hypervisors and explain why the distinction matters in multi-tenant cloud security architecture.
- By the end of this lecture, you will be able to describe the security configuration requirements for hypervisor management interfaces, including isolation, access control, and patching.
- By the end of this lecture, you will be able to explain how CPU speculative execution vulnerabilities such as Spectre and Meltdown created cross-tenant risk in cloud environments and describe the industry response.
- By the end of this lecture, you will be able to describe how VM escape, memory ballooning, and memory scrubbing failures create data exposure risks in virtual environments.
- By the end of this lecture, you will be able to apply three operational disciplines for securing hypervisors, vCPUs, and virtual memory in cloud infrastructure.

Introduction

The moment you run a virtual machine in a cloud environment, you are operating under a security assumption: that the workload on the virtual server next to yours cannot reach your data, your memory, or your processing context. That assumption is enforced by one piece of software, the hypervisor. The hypervisor is simultaneously the most important security boundary in multi-tenant cloud infrastructure and one of the highest-value targets in any cloud environment. Understanding how it works, how it fails, and how to configure it securely is foundational knowledge for any cloud security professional.

This lecture examines the security characteristics of the two hypervisor types, the specific configuration and management practices that maintain hypervisor integrity, and two categories of virtualised hardware that introduce their own security concerns: virtual CPUs and virtual memory. The January 2018 disclosure of Spectre and Meltdown, which affected nearly every modern CPU in production, serves as the most significant real-world demonstration of how vCPU vulnerabilities can create cross-tenant risk at scale in cloud environments. The memory ballooning and VM escape scenarios round out the picture of what happens when the virtualisation boundary fails.

The through line in this lecture is operational discipline: keeping the hypervisor patched and minimally configured, monitoring for anomalous resource usage, and designing workload placement to reduce co-residency risk for the most sensitive data. These are not theoretical controls. They are the specific operational responsibilities of a cloud security practitioner working in or reviewing a cloud environment that runs multiple tenants on shared physical infrastructure.

Core Concepts Explained

Type 1 vs. Type 2 Hypervisors

Definition: A Type 1 hypervisor runs directly on physical hardware without an intervening operating system, minimising attack surface. A Type 2 hypervisor runs on top of a host operating system, which adds a layer of attack surface between the physical hardware and guest VMs.

Business relevance: Multi-tenant cloud environments use Type 1 hypervisors because the security isolation they provide is a prerequisite for running untrusted workloads from different customers on the same physical hardware. A Type 2 hypervisor in production introduces an additional attack surface that can undermine the isolation guarantees customers expect.

Cloud Security Practitioner relevance: A Cloud Security Practitioner evaluating cloud provider architecture or reviewing security questionnaire responses should identify whether the production environment uses Type 1 hypervisors, and flag any use of Type 2 hypervisors in production as a significant risk indicator requiring explanation.

Real-world example: VMware ESXi is a Type 1 hypervisor deployed at scale across enterprise data centres and cloud providers, including in environments hosting HIPAA-regulated healthcare workloads. When a healthcare cloud provider deploys patient data processing workloads on ESXi hosts, the absence of a general-purpose host operating system between the hypervisor and the physical hardware means that even if an attacker gained root-level access to one guest VM, they would need to exploit a vulnerability specifically in the ESXi hypervisor code to move to the host level. That code is purpose-built, minimised, and regularly audited. Compare this to a hypothetical deployment using Oracle VirtualBox on top of Ubuntu Server: an attacker who compromised the Ubuntu host OS through any vulnerability in the host's software, a misconfigured web service, an unpatched library, or a privilege escalation in any host-level process, could potentially reach the VirtualBox hypervisor process and all guest VMs running under it. A CCSP exam question presenting a multi-tenant cloud platform using Type 2 hypervisors in production should immediately be identified as a security architecture concern.

Risks or limitations: The risk is not simply theoretical. If a production environment uses Type 2 hypervisors and the host OS is compromised through any of its many potential attack vectors, every guest VM on that host is exposed. The best practice is to use Type 1 hypervisors for all production multi-tenant deployments and to treat any Type 2 hypervisor in a production context as a finding requiring remediation.

Hypervisor Management Interface Security

Definition: The hypervisor management interface, including web consoles, API endpoints, and administrative CLIs, is the control plane for all virtual machines on a host. It must be isolated on a dedicated management network, protected by role-based access control, and maintained with current patches.

Business relevance: Access to the hypervisor management interface grants control over all guest VMs on the host. Compromise of this interface is equivalent to compromise of every workload the hypervisor runs, making it among the highest-value targets in any cloud environment.

Cloud Security Practitioner relevance: A Cloud Security Practitioner is responsible for ensuring that hypervisor management access is restricted to a defined administrator population, accessible only through controlled network paths, and that all access is logged and monitored for anomalous behaviour.

Real-world example: A cloud security team at a managed hosting provider discovered during a penetration test in 2021 that the VMware ESXi management interface on their production hosts was reachable from the same network segment as the guest VMs, not from a dedicated management VLAN as required by their security policy. The penetration tester, working from a guest VM with no special privileges, was able to reach the ESXi

management interface on port 443 and identify the management software version. Although they did not exploit a vulnerability in the test, the finding was critical: if a guest VM was ever compromised through an application vulnerability, the attacker would immediately have network access to the most powerful control surface in the environment. The remediation involved reconfiguring all ESXi hosts to bind the management interface exclusively to the management VLAN, which was accessible only from the administrator jump host over a VPN-gated connection. After the change, a compromised guest VM had no network path to the hypervisor management interface at all.

Risks or limitations: Role-based access control without network isolation is insufficient. An attacker who can reach the management interface from a compromised guest VM may be able to exploit a vulnerability even without valid credentials if a suitable CVE exists. Network isolation is the defence in depth control that limits management interface exposure when RBAC alone is bypassed.

vCPU Security and CPU Side-Channel Attacks

Definition: Virtual CPUs are virtualised representations of physical processor resources, scheduled onto physical CPU cores by the hypervisor. CPU microarchitectural vulnerabilities, particularly those exploiting speculative execution, can allow cross-tenant data exposure through the shared physical CPU.

Business relevance: CPU-level vulnerabilities in cloud environments are uniquely consequential because they can affect every tenant on a shared physical host simultaneously, are difficult to detect, and require coordinated responses across cloud provider infrastructure, hardware vendors, and operating system maintainers.

Cloud Security Practitioner relevance: A Cloud Security Practitioner must ensure that hypervisor firmware and CPU microcode are current, that vendor-supplied speculative execution mitigations are applied, and that for the most sensitive workloads, dedicated host options that eliminate co-residency are evaluated and used where appropriate.

Real-world example: In January 2018, Google Project Zero researchers disclosed Spectre (CVE-2017-5753 and CVE-2017-5715) and Meltdown (CVE-2017-5754), two classes of vulnerability affecting virtually every Intel, AMD, and ARM processor manufactured in the past two decades. Meltdown exploited a race condition in speculative execution to allow a user-space process to read kernel memory it should never have access to. Spectre was broader: it exploited branch prediction to allow a process to read memory belonging to other processes, including other tenants' VMs on the same physical host. AWS, Microsoft Azure, and Google Cloud Platform each issued emergency communications and began restarting hypervisor hosts within days of the disclosure to apply microcode updates and kernel patches. AWS completed the majority of its infrastructure restart in under a week, affecting customers through brief reboots of EC2 instances. The scale of the remediation operation, affecting millions of virtual machines across three major cloud providers simultaneously, illustrated how a single CPU microarchitectural vulnerability can create an operational emergency for the entire cloud industry. The mitigations applied were a combination of CPU microcode updates from Intel and AMD, kernel page-table isolation patches in Linux and Windows that prevented Meltdown-class reads, and hypervisor changes that mitigated Spectre-class cross-VM attacks.

Risks or limitations: The primary ongoing risk is that new CPU side-channel vulnerabilities continue to be discovered. MDS (Microarchitectural Data Sampling), L1TF (L1 Terminal Fault), and subsequent related vulnerability classes were disclosed in 2018 and 2019. Maintaining current CPU microcode and hypervisor patches is not a one-time action; it is a continuous operational requirement. For workloads processing data classified as requiring exclusive physical CPU access, dedicated hosts that guarantee no co-residency are the correct architectural choice.

Virtual Memory Security: VM Escape and Memory Ballooning

Definition: Virtual memory isolation is the hypervisor's enforcement of the boundary between each VM's allocated memory and all other VMs on the same host. VM escape is the worst-case failure of this boundary. Memory ballooning is a performance optimisation that introduces a risk of cross-VM data exposure if reclaimed memory is not scrubbed before reallocation.

Business relevance: Memory isolation failure in a multi-tenant environment can expose any data that is currently being processed, including decrypted sensitive data that cannot be protected by disk encryption. This is among the most serious categories of cloud infrastructure vulnerability.

Cloud Security Practitioner relevance: A Cloud Security Practitioner must understand that encryption at rest does not protect data while it is in active use in VM memory, and that hypervisor-level memory management requires current patches and correct configuration to prevent cross-VM data exposure.

Real-world example: Memory ballooning is a hypervisor feature used when a host is under memory pressure. The hypervisor inflates a balloon driver inside a guest VM, causing the guest operating system to release memory pages back to the hypervisor. The hypervisor then redistributes those pages to other VMs that need more memory. If the hypervisor does not explicitly zero those pages before giving them to a new VM, the new tenant's VM receives memory that may contain data from the previous tenant's applications, database queries, cached credentials, or encryption keys that were in use at the time the memory was released. A correctly configured and patched hypervisor scrubs all reclaimed memory before reallocation. A misconfigured or unpatched hypervisor may not. In the context of the CCSP exam, a question that asks whether encryption at rest prevents data exposure from memory ballooning has a definitive answer: no. Encryption at rest protects data on disk. Data in active use in VM memory is decrypted and unprotected from memory-level access, which is exactly why hypervisor memory management security is a separate and critical control. VM escape vulnerabilities, while rare, are periodically discovered in hypervisor software. CVE-2018-3646 (L1TF) and multiple CVEs in VMware ESXi over the years represent real-world examples where the VM boundary was at risk. Prompt hypervisor patching is the primary mitigation.

Risks or limitations: The misconception that confidentiality depends only on encryption at rest is one of the most common and dangerous in cloud security. Data being actively processed is in plaintext in memory. If the hypervisor memory isolation boundary fails, encryption provides no protection. Confidential computing technologies, which encrypt data even during processing through hardware-isolated execution environments, address this gap for the most sensitive workloads.

Practical Applications

The following scenarios illustrate how hypervisor, vCPU, and virtual memory security principles apply in professional cloud security decisions.

Evaluating a Cloud Provider's Security Architecture

A cloud security architect at a financial institution is completing a vendor security questionnaire for a new cloud provider the organisation is considering for its trading risk analytics platform. One section of the questionnaire asks about the hypervisor architecture used in the provider's multi-tenant compute environment. The provider's response states that production workloads run on VMware ESXi and the management interface is accessible only from a dedicated management VLAN, not from guest VM networks. The architect also confirms through the provider's published security documentation that CPU microcode updates for Spectre and Meltdown mitigations were applied in 2018 and that the provider publishes a policy of applying CPU microcode updates within 30 days of vendor release. She rates the provider's hypervisor security posture as satisfactory for the analytics platform

but flags one additional requirement: the trading risk calculations involve market-sensitive data that the business classifies as requiring dedicated physical compute isolation. She recommends that the platform provision analytics workloads on the provider's dedicated host service, which guarantees no co-residency with other tenants on the physical CPU, rather than on shared multi-tenant compute.

Responding to a Hypervisor Vulnerability Disclosure

A cloud operations security engineer receives a notification from VMware that a critical vulnerability has been published affecting the ESXi version running on forty-three production hosts in the organisation's private cloud. The vulnerability is rated 9.1 on the CVSS scale and is described as a heap overflow in the ESXi management interface that could allow a remote, unauthenticated attacker to execute code in the hypervisor context. The engineer immediately confirms that the management interface is accessible only from the management VLAN, which reduces the exploitable attack surface to internal administrators and jump host users. She raises a P1 change request through the change advisory board, citing the critical severity and the existing network control as a compensating measure during the patch window. The change advisory board approves an emergency maintenance window for that evening. She coordinates with the hypervisor team to apply the patch across all forty-three hosts in a rolling maintenance sequence, with a 30-minute verification window on each host before moving to the next. The management VLAN isolation does not eliminate the risk but provides a meaningful compensating control that allows the team to patch in a controlled sequence rather than an emergency simultaneous restart.

Investigating a Suspected VM Memory Exposure

A security analyst at a managed cloud provider receives a report from a tenant that their application logs show database queries they did not execute being returned in result sets, raising a concern that another tenant's database queries might be visible in their application's responses. The analyst investigates the memory ballooning configuration on the host servers the tenant's VMs are running on and finds that the host cluster is running an older version of the hypervisor that has a known issue with memory page scrubbing under high memory pressure. The scrubbing is configured but not executing correctly when the balloon driver releases pages under a specific memory pressure threshold. The analyst immediately migrates the affected tenant's VMs to a different host cluster running the patched hypervisor version, documents the configuration failure as a security incident, and opens a formal investigation to determine whether any other tenant's data was exposed through the memory page issue. She also initiates a review of the memory scrubbing configuration and version state across all host clusters in the environment, finding two additional clusters running the vulnerable version.

Best Practices

- **Use Type 1 hypervisors for all production multi-tenant deployments without exception.** Type 2 hypervisors introduce an entire general-purpose operating system as an additional attack surface between the physical hardware and guest VMs. In a multi-tenant environment, this additional surface can be exploited to reach all guest VMs without a hypervisor-specific vulnerability. Review any production deployment that uses Type 2 hypervisors as a critical finding requiring architectural remediation.
- **Isolate the hypervisor management interface on a dedicated management VLAN inaccessible from guest VM networks.** Network reachability to the management interface from a guest VM enables an attacker who compromises any workload on the host to attempt hypervisor exploits. Management network isolation is a defence-in-depth control that limits the blast radius of guest VM compromise. Verify isolation with a network reachability test from a guest VM to the management interface IP range as part of each security review.

- **Apply CPU microcode and hypervisor patches for speculative execution vulnerabilities as emergency changes, not standard maintenance cycle patches.** Spectre, Meltdown, and subsequent CPU side-channel vulnerabilities carry cross-tenant data exposure risk that makes them operationally different from typical software vulnerabilities. Establish a policy that classifies CPU microcode updates addressing side-channel vulnerabilities as emergency changes with a maximum patching window of 72 hours after vendor release, not the standard monthly patch cycle.
- **Verify memory scrubbing configuration and test it under load conditions, not just in idle state.** Memory ballooning scrubbing failures typically occur under memory pressure conditions that do not exist during standard configuration testing. Test the scrubbing behaviour by writing a known pattern to memory in a test VM, triggering the balloon driver under memory pressure, verifying that the released pages contain zeroes before reallocation, and confirming this result after each major hypervisor version update.
- **For workloads classified as requiring physical CPU isolation, provision dedicated hosts rather than shared multi-tenant compute.** No software or microcode mitigation provides the same isolation assurance as physical separation of CPU resources. If a workload processes data where cross-tenant CPU-level exposure cannot be accepted, the correct architectural decision is dedicated hosts. Document this classification decision in the workload's risk register and review it when new CPU vulnerability classes are disclosed.

Common Mistakes and Misconceptions

Misconception: Deploying Type 2 hypervisors in production multi-tenant environments, treating them as equivalent to Type 1 for security purposes.

Correct approach: Use Type 1 bare-metal hypervisors for all production multi-tenant deployments. Type 2 hypervisors are appropriate for development and testing environments only, where the host OS attack surface is acceptable.

Misconception: Assuming that encryption at rest protects data from hypervisor memory boundary failures, such as VM escape or memory ballooning data leakage.

Correct approach: Encryption at rest protects data on disk. Data being actively processed in VM memory is in plaintext and is vulnerable to hypervisor-level memory access failures. Hypervisor patching and memory scrubbing configuration are separate, required controls for data-in-use protection.

Misconception: Treating CPU microcode updates for speculative execution vulnerabilities as routine patches to be applied in the next standard maintenance window.

Correct approach: CPU side-channel vulnerabilities with cross-tenant exposure potential should be classified as emergency changes with a defined maximum patching window, reflecting the elevated risk they represent in multi-tenant cloud environments.

Misconception: Leaving the hypervisor management interface reachable from guest VM network segments, relying solely on authentication controls to protect it.

Correct approach: Isolate the hypervisor management interface on a dedicated management VLAN that is not reachable from guest VM networks. Verify isolation with network reachability testing. Network isolation is the defence-in-depth control that limits management interface exposure when authentication controls are bypassed.

checklist: Hypervisor Security Configuration Verification Checklist

Use this checklist to verify the security configuration of hypervisor hosts in a cloud or virtualised environment. Each item represents a specific control that should be verified, not assumed.

- ☐ Hypervisor type is Type 1 (bare-metal) for all production multi-tenant hosts; Type 2 hypervisors are absent from production.
- ☐ Hypervisor management interface is bound to a dedicated management VLAN IP address, not to the same interface used by guest VM traffic.
- ☐ Network reachability test confirms: no path from guest VM networks to the hypervisor management interface IP range.
- ☐ Role-based access control is configured for hypervisor management; administrator roles are defined with minimum required permissions and reviewed quarterly.
- ☐ Hypervisor software version and CPU microcode are current; last update date is within the organisation's defined patching window for the vulnerability severity of available updates.
- ☐ Spectre, Meltdown, and subsequent CPU side-channel mitigations are confirmed applied at the hypervisor, OS kernel, and CPU microcode layers.
- ☐ Memory scrubbing (zeroing of reclaimed pages) is configured and verified as active; last verification test date is documented.
- ☐ Memory ballooning is configured to not exceed thresholds that could trigger scrubbing failures; thresholds are tested under load conditions.
- ☐ VM escape vulnerability CVEs for the installed hypervisor version are reviewed; no unpatched critical VM escape vulnerabilities exist.
- ☐ Hypervisor management access logs are ingested into the centralised SIEM and monitored for anomalous access patterns.
- ☐ Dedicated host configuration is in use for workloads classified as requiring physical CPU isolation; co-residency with other tenants is verified as absent.

How to use this: Complete this checklist on initial host commissioning, after each major hypervisor version update, and as part of each annual cloud security review. Items marked as not in place should be escalated as findings with defined remediation timelines.

Knowledge Check

Test your understanding before moving on. Answers are shown below each question.

Q1. A cloud architect is reviewing the security architecture of a managed cloud platform that offers virtual machine hosting to multiple customers. The documentation states that production workloads run on VMware Workstation installed on Ubuntu Server hosts. What is the most significant security concern with this architecture?

- A) A. VMware Workstation does not support VLANs for network segmentation between tenant VMs.

- B) B. Ubuntu Server does not support hardware security modules, making cryptographic key protection impossible.
- C) C. A Type 2 hypervisor runs on top of a general-purpose host OS, meaning that any compromise of the Ubuntu host OS could expose all guest VMs on that host.
- D) D. VMware Workstation cannot be patched for CPU microcode vulnerabilities without a host OS reboot.

Answer: C. A Type 2 hypervisor runs on top of a general-purpose host OS, meaning that any compromise of the Ubuntu host OS could expose all guest VMs on that host.

Q2. In January 2018, the Spectre and Meltdown vulnerabilities were disclosed, affecting nearly all modern CPUs. Which statement most accurately describes the security implication for multi-tenant cloud environments?

- A) A. The vulnerabilities allowed attackers to decrypt data stored on encrypted disks within a virtual machine.
- B) B. The vulnerabilities exploited speculative execution to potentially allow one tenant's VM to read memory belonging to another tenant's VM on the same physical host.
- C) C. The vulnerabilities allowed attackers to escape from a guest VM to the hypervisor layer and gain hypervisor-level code execution.
- D) D. The vulnerabilities affected only Windows-based virtual machines and did not impact Linux or cloud-native workloads.

Answer: B. The vulnerabilities exploited speculative execution to potentially allow one tenant's VM to read memory belonging to another tenant's VM on the same physical host.

Q3. A cloud environment is experiencing high memory pressure on its virtualised hosts. The hypervisor uses memory ballooning to reclaim memory from lightly loaded VMs and redistribute it to others. Which of the following represents the most significant security risk of this configuration if the hypervisor is not correctly patched?

- A) A. The ballooning process may consume too much CPU time, reducing performance for all tenant VMs.
- B) B. The balloon driver may fail to release memory back to the hypervisor when pressure reduces, causing memory fragmentation.
- C) C. Memory pages released through ballooning may not be zeroed before reallocation, potentially exposing one tenant's data to another tenant's VM.
- D) D. Ballooning may cause the hypervisor management interface to become unresponsive, requiring a host restart.

Answer: C. Memory pages released through ballooning may not be zeroed before reallocation, potentially exposing one tenant's data to another tenant's VM.

Q4. A security engineer is reviewing the hypervisor configuration on a private cloud host and discovers that the ESXi management interface is accessible from the same network segment as the guest VMs running on that host. Which risk does this configuration most directly create?

- A) A. Guest VMs may be able to reach the hypervisor management interface and attempt exploitation of hypervisor vulnerabilities, even without guest OS compromise.
- B) B. Network performance for guest VMs will be degraded because management traffic competes with workload traffic on the same interface.
- C) C. The ESXi hypervisor cannot enforce memory isolation between VMs when the management interface is on a shared network segment.

D) D. Certificate validation for the management interface will fail because the guest VM network does not trust the management console's certificate authority.

Answer: A. Guest VMs may be able to reach the hypervisor management interface and attempt exploitation of hypervisor vulnerabilities, even without guest OS compromise.

Q5. Why does encryption of data at rest not protect against VM memory boundary failures such as a hypervisor memory scrubbing defect?

A) A. Encryption at rest uses algorithms that are only effective for protecting data on physical storage media, not virtual disk files.

B) B. Data that is being actively processed is decrypted in VM memory; encryption at rest only protects data stored on disk, not data in active use.

C) C. Virtual disk encryption is handled by the hypervisor, which has access to all memory anyway.

D) D. Encryption at rest keys are stored in the same memory that could be exposed, so the protection is circular.

Answer: B. Data that is being actively processed is decrypted in VM memory; encryption at rest only protects data stored on disk, not data in active use.

Reflection Questions

1. Consider a cloud environment you work with or have studied. Is it using a Type 1 hypervisor? If you do not know, what would you ask the cloud provider to verify this, and what documentation would you expect to receive?
2. The Spectre and Meltdown disclosures in 2018 required cloud providers to restart significant portions of their infrastructure within days. If a similar disclosure happened tomorrow, what is your organisation's process for evaluating the severity of a CPU microarchitectural vulnerability and determining whether it requires emergency response?
3. Memory scrubbing is a configuration item that most security practitioners rarely think about explicitly. How would you verify that memory scrubbing is functioning correctly in a virtualised environment you are responsible for securing?
4. If a critical VM escape vulnerability were disclosed in the hypervisor version your organisation runs, what is the current maximum time before that patch would be applied? Is that window acceptable given the potential for full host compromise through a guest VM?
5. Confidential computing technologies encrypt data during processing, addressing the in-memory exposure gap that encryption at rest cannot fill. Is there any workload in your environment where this additional protection would be warranted?

Key Takeaways

- Type 1 hypervisors run directly on physical hardware with minimal attack surface; Type 2 hypervisors run on a host OS, adding a full general-purpose attack surface that is inappropriate for production multi-tenant environments.
- Hypervisor management interfaces must be isolated on dedicated management networks with no routing path from guest VM segments; network isolation is the defence-in-depth control that supplements authentication.

- Spectre and Meltdown, disclosed in January 2018, exploited speculative execution to create potential cross-tenant memory reads through the shared physical CPU in cloud environments, requiring coordinated emergency response across cloud providers and hardware vendors.
- VM escape vulnerabilities exploit hypervisor software bugs to break out of the VM boundary entirely; memory ballooning defects can expose one tenant's in-memory data to another if pages are not zeroed before reallocation.
- Encryption at rest does not protect data being actively processed in VM memory; in-memory data is decrypted and vulnerable to hypervisor-level boundary failures.
- Three operational disciplines govern this area: keep the hypervisor updated and minimally configured with its management interface isolated, monitor for anomalous CPU and memory usage, and use dedicated hosts for workloads requiring physical CPU isolation.
- CPU microcode updates for speculative execution vulnerabilities should be treated as emergency changes, not standard maintenance cycle patches, due to their cross-tenant exposure potential.

Recommended Next Actions

Put this lecture to work within the next seven days:

6. Confirm the hypervisor type in use in a cloud environment you have access to and verify that the management interface is accessible only from a dedicated management network, not from guest VM segments.
7. Review the current hypervisor and CPU microcode patch levels in your environment and compare them against the vendor's latest published versions; document any gaps and their current CVE severity ratings.
8. Identify one workload in your environment that processes data you would classify as requiring physical CPU isolation and investigate whether your cloud provider offers a dedicated host option for that workload.
9. Test or request evidence of memory scrubbing verification for the hypervisor hosts in your environment; document the last test date and the methodology used.